

S1Q13

Question 13 – Stage 1

I have a stopwatch that records time in the form $HH : MM : SS$.

It resets to $00 : 00 : 00$ after 100 hours.

During this 100 hour interval, how many times will the display use precisely two different digits (like $00 : 00 : 01$ and $99 : 59 : 59$)?

Working

```
In [1]: from itertools import product

twoDigitCount = 0

for time in product(range(100), range(60), range(60)):
    time = [str(t) for t in time]
    time = "".join([t if len(t) == 2 else "0"+t for t in time])

    twoDigitCount += 1 if len({digit for digit in time}) == 2 else 0 # Convert to set to disallow repeats

print(f"Count: {twoDigitCount}")

Count: 1290
```